

Kelsey McGuire

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Professional Summary

MSc Geography and BSc Geographical Sciences graduate from the University of British Columbia with 3+ years of experience with aquatic and terrestrial environmental data collection, data cleaning, quality assurance, and R, ArcGIS and QGIS programs. Proven ability to coordinate field campaigns, manage complex ecological datasets, and translate technical findings into stakeholder-focused outreach and knowledge products.

Technical Skills & Certifications

Field Instrumentation: Hanna & YSI Water Sondes, Water Grab Sampling, LICOR Trace Gas Analyzers, Soil Coring

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Online, QGIS, ENVI

Software & Programming: R (tidyverse, ggplot2, ShinyApp), Git/GitHub, Python, Microsoft Suite

Certifications: Remote/Workplace First Aid, Transport Endorsement, Safe Kayaking, Class 5 License, DELF B2

Education

M. Sc., GEOGRAPHY

2024 - 2026

University of British Columbia (Vancouver, BC)

Thesis Title: *A Little Sedge Goes a Long Way: Vegetation as a Key Driver of Littoral Methane Flux*

Thesis Advisor(s): *Dr. McKenzie Kuhn*

B. Sc., GEOGRAPHICAL SCIENCES WITH DISTINCTION

2021 - 2024

University of British Columbia (Vancouver, BC)

Relevant Work Experience

GRADUATE RESEARCHER | BOREAL-ARCTIC BIOGEOCHEMISTRY LAB, UBC | 09/2025 –

- Collected, processed, and analyzed over 300 ecological and water quality samples using standardized sampling techniques such as grab samples, Van Dorn sampling, and carbon accounting through manual chambers.
- Developed reproducible data processing workflows using R and GitHub to automate data cleaning, multi-source dataset integration, quality assurance/quality control (QA/QC), statistical analyses, and visualizations.
- Produced technical reports, grant proposals, plain-language knowledge products, a scientific publication, and outreach materials for community partners, grant agencies, and academic audiences.

NSERC USRA STUDENT RESEARCHER | GEOSPATIAL ECOLOGY LAB, UBC | 05/2024– 08/2024

- Collected and processed 499 soil samples across six savanna–forest mosaic sites using standardized field and laboratory protocols with post-processing for quality control and analysis done using R and QGIS.
- Managed spatial and ecological datasets, ensuring data accuracy, consistency, and documentation throughout field and laboratory workflows.

WORKLEARN STUDENT | GEOSPATIAL ECOLOGY LAB, UBC | 09/2023– 04/2024

- Developed and tested Google Earth Engine workflows to identify fire occurrence across Cambodia and assess relationships between fire frequency and land cover classifications.
- Designed and maintained an open-source laboratory website and internal knowledge wiki using R, Git, and GitHub to improve research data organization and accessibility.
- Supported collaborative coding and documentation practices through version control and shared repository management.

UNDERGRADUATE RESEARCH ASSISTANT | UBC MICROMETEOROLOGY LAB, UBC | 05/2023 – 03/2024

- Conducted greenhouse gas field measurements at 48 sampling sites within a post-fire peatland ecosystem using standardized field protocols.
- Developed automated R workflows for processing phenology datasets from international monitoring sites and collaborated on an open-source Shiny application for interactive environmental data visualization.

Relevant Courses

FRST505 | CLIMATE CHANGE COMMUNICATION AND ENGAGEMENT | UNIVERSITY OF BRITISH COLUMBIA

- Discussed how climate change has historically been communicated, the barriers to proper communication such as misinformation, and how to best engage communities with conversations on climate change impacts and local responsibilities.
- Key projects included small and large group discussions on climate change and individual relationships to the issue and a board game with the goal of creating conversations about fossil fuel use and what needs to change.

STAT545 | INTRODUCTORY DATA ANALYSIS | UNIVERSITY OF BRITISH COLUMBIA

- Focused on efficiently employing functions within R's tidyverse and ggplot2 packages. Main tasks included learning to properly filter, clean, and present data based on the structure and goals of the given objective.
- Key outputs included a shared GitHub project that employed R Markdowns and displayed the use of many tidyverse functions including filter(), select(), mutate(), alongside a Shiny App that was created to visualize methane data against environmental variables based on a user supplied dataset.

GEOS370 (ADVANCED GEOGRAPHIC INFORMATION SCIENCE); GEOS373 (INTRODUCTORY REMOTE SENSING)

- Designed a multi-criteria habitat suitability model in ArcGIS Pro using Suitability Modeler, Kriging, and Nearest Neighbour analyses to identify potential caribou habitat in northern Canada. Published project outputs through ArcGIS Online StoryMaps and automated portions of the workflow using Python.
- Produced both static and web-based maps using ArcGIS Pro and ArcGIS Online to communicate spatial analyses to technical and non-technical audiences.
- Performed satellite image preprocessing, classification, and change detection in ENVI and Google Earth Engine to compare supervised and unsupervised classification methods for assessing coastal change around New Orleans, Louisiana.

Scholarships & Awards *(totaling over \$90,000)*

CGU BEST STUDENT POSTER – BIOGEOSCIENCES	MAY 2026
WALL RESEARCH AWARD (DECLINED)	\$25 000 (JUNE 2025)
NORTHERN SCIENTIFIC TRAINING PROGRAM	\$3300 (MAY 2025)
NSERC CANADA GRADUATE SCHOLARSHIP – MASTER'S (CGS-M)	\$27 000 (APR. 2025)
WCS WESTON FAMILY BOREAL RESEARCH FELLOWSHIP	\$10 000 (MAR. 2025)
BRITISH COLUMBIA GRADUATE SCHOLARSHIP	\$17 500 (MAY 2024)
NSERC UNDERGRADUATE STUDENT RESEARCH AWARD (USRA)	\$6000 (MAY 2024)
UBC GEOGRAPHY ALUMNI SCHOLARSHIP	\$3450 (NOV. 2023)

Memberships & Extracurriculars

ASLO STUDENT MEMBER	2025/2026
CGU STUDENT MEMBER	2025/2026
UBC GEOGRAPHY GRADUATE ASSOCIATION	2024/2025, 2025/2026
UBC GEOGRAPHY STUDENTS' ASSOCIATION – VP EXTERNAL	2023/2024